Lecture 10

- The Legendre Polynomials (Example of GS process:)
- Orthogonal Projections
- Orthogonal Least Squares
- Orthogonal Subspaces (Fund. MAT. Subspaces)

Last time we discussed orthonormal bases in abstract inner product spaces, we noticed their importance (it is easy to represent other vectors in terms of basis vectors, because it is easy to compute coordinates), and we learned how to construct orthonormal bases using the Gramm-Schidt process.

Orthonormal bases play especially <u>important role</u> in functional spaces.

- approximation theory
- interpolation theory
- Fourier analysis
- solutions of PDEs

However, unlike $\mathbb{R}^n$, most obvious bases of typical functional spaces are not orthogonal w.r.t. any natural inner product.

<u>Example</u>: Consider $P^{(2)}$, the vector space consisting of all quadratic polynomials, on $[-1,1]$, equipped with the L^2 inner product $\langle p, q\rangle = \int_{-1}^{1} p(x)q(x)\,dx$.

The standard monomials $1, x, x^2$ do form a basis, but they are not orthogonal

$$\langle 1, x\rangle = \int_{-1}^{1} x\,dx = 0, \quad \langle 1, x^2\rangle = \frac{2}{3}, \quad \langle x, x^2\rangle = 0 \qquad \square$$

Thus, we need to use the GS process (alg.) to orthogonalize them.

<u>Remark</u>: $[-1,1]$ is for definiteness.

Legendre Polynomials

Let's construct an orthonormal basis for the inner product space $\left(P^{(n)}, \langle\cdot,\cdot\rangle_{L^2}\right)$. (polynom on $[-1,1]$ of deg $\le n$)

Let's apply the GS orthogonalization process to the elementary, but not orthogonal monomial basis $1, x, \dots, x^n$, and denote $q_0(x), \dots, q_n(x)$ the resulting orthogonal polynomials. Let's compute first few q_i

<u>Recall GS</u>: $\{w_i\} \to \{v_i\}$

$$v_k = w_k - \sum_{i=0}^{k-1} \frac{\langle w_k, v_i\rangle}{\|v_i\|^2} v_i$$

- $q_0(x) = 1$
- $q_1(x) = x - \dfrac{\langle x, 1\rangle}{\|1\|^2} 1 = x$
- $q_2(x) = x^2 - \dfrac{\langle x^2, 1\rangle}{\|1\|^2} 1 - \dfrac{\langle x^2, x\rangle}{\|x\|^2} x = x^2 - \dfrac{\frac{2}{3}}{2} = x^2 - \frac{1}{3}$

In general,

$$(*) \quad \boxed{q_k(x) = x^k - \sum_{i=0}^{k-1} \frac{\langle x^k, q_i(x)\rangle}{\|q_i(x)\|^2} q_i(x)} \qquad k = 0, \dots, n$$

monic Legendre polynomials (leading coefficient is 1)

Legendre introduced them in 18th centure and used them for studying Newtonian gravitation

Since $q_0, \dots, q_n$ form a basis in $\mathcal{P}^{(n)}$, any polynomial $p(x)$ of deg $= n$ can be uniquely written as a linear combination $p(x) = \alpha_0 q_0(x) + \dots + \alpha_n q_n(x)$.

The coefficients are simply (Th 2, lecture 9): $\alpha_i = \dfrac{\langle p, q_i \rangle}{\|q_i\|^2} = \dfrac{\int_{-1}^{1} p(x) q_i(x)\,dx}{\int_{-1}^{1} q_i^2(x)\,dx}$.

Def The classical Legendre polynomials are rescaled monic Legendre polynomials, $P_k(x) = \alpha_k q_k(x)$, such that $P_k(1) = 1$.

It can be shown that

$$\boxed{P_k(x) = \frac{(2k)!}{2^k (k!)^2} q_k(x)} \qquad k = 0, \dots, n$$

not orthonormal HW: find the norm Ex 5.4.9

for our purposes does not matter, but used in some applications

Formula (*) allows to compute the Legendre polynomials recursively. It turns out there is an explicit formula.

Theorem (O. Rodrigues) : portuguese

$$\boxed{P_k(x) = \frac{1}{2^k k!} \underbrace{\frac{d}{dx^k}(x^2-1)^k}}$$

Proof is technical, not interesting, in the text book.

poly of deg $= k$, which is $\perp$ to all $1, x, \dots, x^{k-1}$ (simple check using integration by parts)

$\Leftarrow \dfrac{d}{dx^k}(x^2-1)^k \propto q_k(x) \propto P_k(x) \Leftarrow$

find const. of propor. by comparing the leading terms

Q: How to construct an orthogonal basis for $\mathcal{P}^{(n)}$ on $[a,b]$?

Idea: Linear change of variables

$$x = \frac{2y-b-a}{b-a} \quad \text{maps} \quad y \in [a,b] \leftrightarrow x \in [-1,1]$$

Remark Of course, we can use the GS algorithm "from scratch". but there is a better idea.

- This map induces a one-to-one map

$$\begin{array}{ccc} \mathcal{P}^{(n)}_{[a,b]} & \longleftrightarrow & \mathcal{P}^{(n)}_{[-1,1]} \\ \rotatebox{90}{$\in$} & & \rotatebox{90}{$\in$} \\ F & \longleftrightarrow & f \end{array} \qquad \begin{aligned} F(y) &= f\left(\frac{2y-b-a}{b-a}\right) \\ f(x) &= F\left(\frac{a+b+(b-a)x}{2}\right) \end{aligned}$$

- How are the inner products related?

$$\langle F, G \rangle_{L^2[a,b]} = \int_a^b F(y) G(y)\,dy = \left[y = \frac{a+b+(b-a)x}{2} \right] = \int_{-1}^{1} f(x) g(x) \frac{b-a}{2}\,dx = \frac{b-a}{2} \langle f, g \rangle_{L^2[-1,1]}$$

Orthogonality is preserved under this change of variables

Theorem The transformed Legendre polynomials

$$\tilde{P}_k(y) = P_k\left(\frac{2y-b-a}{b-a}\right)$$

form an orthogonal basis for $\mathcal{P}^{(n)}$ on $[a,b]$ w.r.t. $\langle \cdot, \cdot \rangle_{L^2}$

Orthogonal Projections

→ Geometric interpretation of G.S. process
→ Geometric interpretation of Least Squares

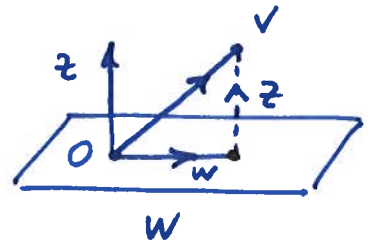
Let V be an inner product space (could be ∞-dim), and $W \subset V$ its subspace (finite dim.).

Def A vector $z \in V$ is said to be orthogonal to the subspace $W \subset V$ if z is orthogonal to every $w \in W$: $\langle z, w \rangle = 0 \quad \forall w \in W$.

Remark A working example: $(\mathbb{R}^n, \cdot)$

If $w_1 \dots w_n$ is a basis for W, then $z \perp W \iff z \perp w_i$, $i=1,\dots n$.

Def The orthogonal projection of $v \in V$ onto W is the element $w \in W$ such that $z = v - w \perp W$, $w = \mathrm{proj}_W^{\perp} v$.



Orthogonal projections of vectors can be easily found if we use an orthonormal basis in W.

Theorem Let $u_1 \dots u_n$ be an orthonormal basis for W. Then the orthogonal projection of $v \in V$ onto W is

$$\boxed{w = \mathrm{proj}_W v = \sum_{i=1}^{n} \alpha_i u_i, \quad \alpha_i = \langle v, u_i \rangle}$$

Proof Since $u_1 \dots u_n$ is a basis of W, $w = \sum_{i=1}^{n} \alpha_i u_i$

Since $(v-w) \perp W$, $(v-w) \perp u_i$ for all i

$$\Rightarrow \quad 0 = \langle v-w, u_i \rangle = \langle v, u_i \rangle - \langle \sum_{j=1}^{n} \alpha_j u_j, u_i \rangle = \langle v, u_i \rangle - \alpha_i \qquad \square$$

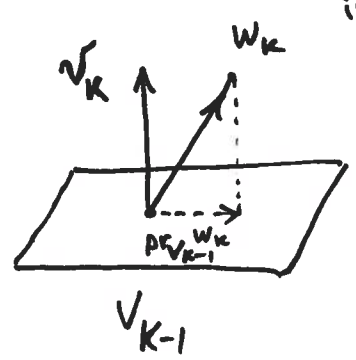
Remark 1 If $v_1 \dots v_n$ is orthogonal basis for W (not necesarily orthonormal) then $\mathrm{proj}_W v = \sum_{i=1}^{n} \alpha_i v_i$, $\alpha_i = \dfrac{\langle v, v_i \rangle}{\|v_i\|^2}$.

Remark 2 Expressions for α_i exactly coincide with the coefficients of a vector $v \in V$ in the orthonormal/orthogonal basis $u_1 \dots u_n$ for V.
This is not surprising: if $v \in W \Rightarrow \mathrm{proj}_W v \equiv v$.

Remark 3 Recall GS again: $\{w_i\} \to \{v_i\}$

$$v_k = w_k - \underbrace{\sum_{i=1}^{k-1} \frac{\langle w_k, v_i \rangle}{\|v_i\|^2} v_i}_{\mathrm{proj}_{V_{k-1}} w_k}, \quad V_{k-1} = \mathrm{span}\{v_1 \dots v_{k-1}\} = \mathrm{span}\{w_1 \dots w_{k-1}\}$$

Geometric interpretation of G.S.

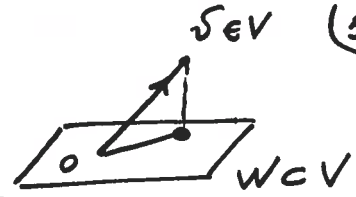


basis of V_{k-1} $\{w_1 \dots w_{k-1}\}$

$\{v_1 \dots v_{k-1}\}$ orthog. basis of V_{k-1}

Orthogonal Least Squares

As the picture suggests, orthogonal projection of a vector onto a subspace, is exactly the least squares vector. (the closest point in the subspace)



Theorem Let $W \subset V$ be a finite-dimensional subspace of an inner product space V, and $v \in V$. Then $\text{proj}_W v$ is the closest point to v in W (the least squares minimizer)

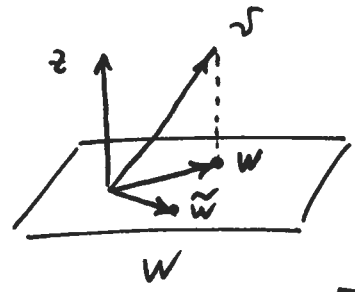
Proof Let $w = \text{pr}_W v$, and $z = v - w$

$z \perp W$

Then for any $\tilde{w} \in W$

$$\|v - \tilde{w}\|^2 = \|z + \underbrace{w - \tilde{w}}\|^2 = \|z\|^2 + \|w - \tilde{w}\|^2 + 2\underbrace{\langle z, w - \tilde{w}\rangle}_{0}$$

$$\Rightarrow \min_{\tilde{w} \in W} \|v - \tilde{w}\| = \|z\| \text{ and it is achieved at } \tilde{w} = w.$$



$\square$

~~This gives an alternative way for finding the least squares solutions of linear systems. So, suppose we want to find the least squares sol. of $Ax = b$. Assume that $\ker A = \{0\}$.~~

~~Alternative Method:~~
- ~~more numerically stable~~
- ~~more efficient~~

~~$\Rightarrow x^* = (A^T A)^{-1} A^T b$~~

This gives an alternative way to compute the closest point.

Recall: If $w_1 \dots w_n$ is a basis for $W \subset V$, (lecture 7) and $v \in V$. Then the closest to v point in W is $w = \sum_{i=1}^{n} \alpha_i w_i$, where $\alpha = K^{-1} f$, $K = (\langle w_i, w_j \rangle)$, $f_i = \langle w_i, v \rangle$

Alternative Method
- more efficient
- more numerically stable

① GS: $\{w_i\} \to \{v_i\}$ ← orthonormal basis of W

② $w = \text{proj}_W(v)$

$$= \sum_{i=1}^{n} \alpha_i v_i, \quad \alpha_i = \langle v, v_i \rangle$$

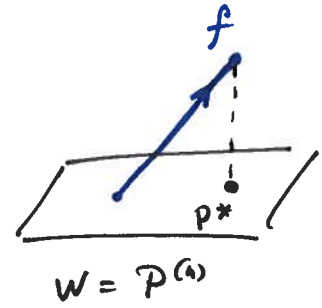
Application to function Approximation

Suppose $f(x)$ is a function on $[-1, 1]$ and we want to approximate it by a polynomial $p^*(x) \in P^{(n)}$. So we want to find

$$p^*(x) = \arg\min_{p \in P^{(n)}} \|f - p\|_{L^2}$$

Legendre polynomials

Geometrically: $p^* = \text{proj}_W f = \sum_{i=0}^{n} \alpha_i P_i(x)$,



$$\text{where } \alpha_i = \frac{\langle f, P_i \rangle}{\|P_i\|^2} = \frac{2i+1}{2} \int_{-1}^{1} f(x) P_i(x)\, dx$$

(This approach is better)

Orthogonal Subspaces

We can naturally extend the notion of orthogonality from vectors to subspaces

Def Two subspaces $U, W \subset V$ are orthogonal ($U \perp W$) if every vector in U is orthogonal to every vector in W.

$$\langle u, w \rangle = 0 \qquad \forall u \in U,\ w \in W$$

Obviously, $U \perp W \iff \langle u_i, w_j \rangle = 0 \ \forall i, j$, where $u_1 \dots u_k$ is a basis in U, $w_1 \dots w_m$ is a basis in W.

Def The orthogonal compliment to a subspace $W \subset V$, denoted $W^\perp$, is the set of all vectors in V which are orthogonal to W

$$W^\perp = \{ v \in V : \langle v, w \rangle = 0 \ \forall w \in W \}$$

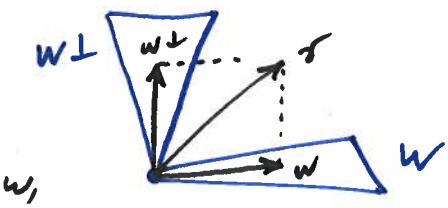
Remark: • $W^\perp$ is a subspace : [redacted] $\langle \alpha v_1 + \beta v_2, w \rangle = \alpha \langle v_1, w \rangle + \beta \langle v_2, w \rangle = 0$.

• Moreover, $W \cap W^\perp = \{0\}$: if $w \in W \cap W^\perp \Rightarrow \langle \underset{W}{\underset{\cap}{w}}, \underset{W^\perp}{\underset{\cap}{w}} \rangle = 0 \Rightarrow w = 0$.

Remark $W^\perp$ depends on $\langle \cdot, \cdot \rangle$!

Theorem: Let $W \subset V$ be a finite-dimensional supspace of an inner product space. Then $\forall v \in V \ \exists!\ w \in W$ and $w^\perp \in W^\perp$ such that $v = w + w^\perp$

Proof [redacted]



$\exists$: let $w = \mathrm{proj}_W(v) \Rightarrow \overbrace{v - w}^{w^\perp} \perp W \Rightarrow w^\perp \in W^\perp$

$!$: $\left.\begin{array}{l} v = w_1 + w_1^\perp \\ v = w_2 + w_2^\perp \end{array}\right\} \Rightarrow \overbrace{\underbrace{w_2 - w_1}_{\in W}}^{0} = \overbrace{\underbrace{w_2^\perp - w_1^\perp}_{\in W^\perp}}^{0} \Rightarrow \begin{array}{l} w_2 = w_1 \\ w_2^\perp = w_1^\perp \end{array}$

$\square$

Corollary: $\dim V = \dim W + \dim W^\perp$ (if $w_1 \dots w_m$ is a basis in W and $w_1^\perp \dots w_k^\perp$ is a basis in $W^\perp$ $\Rightarrow \{w_i\} \cup \{w_j^\perp\}$ is a basis in V)

Corollary: $(W^\perp)^\perp = W$

($\forall v \in V\ \exists!\ w^\perp \in W^\perp$ and $(w^\perp)^\perp \in (W^\perp)^\perp$: $V = \underset{W^\perp}{\underset{\cap}{w^\perp}} + \underset{(W^\perp)^\perp}{\underset{\cap}{(w^\perp)^\perp}} \Rightarrow (w^\perp)^\perp \in W$.

Remark This is true only for <u>finite-dim</u> subspaces W !

See p. 270 [OS] for an example. $\qquad (W^\perp)^\perp \supseteq W$

[redacted]

- Orthogonality of the Fundamental Matrix Subspaces

[redacted]

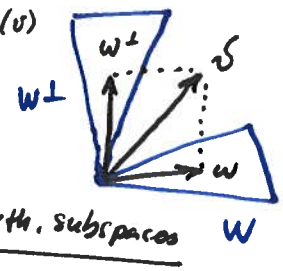
- ~~Geometry of $A: \mathbb{R}^n \to \mathbb{R}^m$~~
- ~~Eigenvalues~~ (remainder of what should be known) ~~& Eigen vectors~~

Last time we finished with the notion of orthogonal subspaces and we discussed their basic properties, namely:

① $\forall v \in V \ \exists! \, w \in W$ and $w^\perp \in W^\perp$: $v = \underbrace{w}_{\text{proj}_W(v)} + \underbrace{w^\perp}_{\text{proj}_{W^\perp}(v)}$

② $\dim V = \dim W + \dim W^\perp$

③ $(W^\perp)^\perp = W$



Fundamental Subspaces associated with a matrix provide an important example of orth. subspaces

Recall: $\mathbb{M}_{m\times n} \ni A = [a_1|\cdots|a_n] = \begin{bmatrix} a^1 \\ a^2 \\ \vdots \\ a^m \end{bmatrix}$

$\ker(A) = \{x \in \mathbb{R}^n : Ax = 0\} \subset \mathbb{R}^n$

$\text{im}(A) = \{Ax : x \in \mathbb{R}^n\} = \text{span}\{a_1, \dots, a_n\} \subset \mathbb{R}^m$

$\text{coker}(A) = \ker(A^T) = \{y \in \mathbb{R}^m : A^T y = 0\} \subset \mathbb{R}^m$

$\text{coim}(A) = \text{im}(A^T) = \{A^T y : y \in \mathbb{R}^m\} = \text{span}\{a^1, \dots, a^m\} \subset \mathbb{R}^n$

Fundamental Theorem (Part I) (where $r = \text{rank}(A)$):

$$\underbrace{\begin{cases} \dim \ker(A) = n - r \\ \dim \text{coim}(A) = r \end{cases} \qquad \begin{cases} \dim \text{im}(A) = r \\ \dim \text{coker}(A) = m - r \end{cases}}$$

+ ② : are these subspaces orthogonal?

Fundamental Theorem (Part II)

Let A be any $m\times n$ matrix.

1. $\ker(A)$ and $\text{coim}(A)$ are orthogonal compliments in $(\mathbb{R}^n, \cdot)$, dot product
2. $\text{im}(A)$ and $\text{coker}(A)$ are orthogonal compliments in $(\mathbb{R}^m, \cdot)$,

$$\boxed{\begin{aligned} \ker(A) &= \text{coim}(A)^\perp \\ \text{coker}(A) &= \text{im}(A)^\perp \end{aligned}}$$

Proof Let's prove that $\ker(A) = \text{coim}(A)^\perp$. (second part will follow automatically)

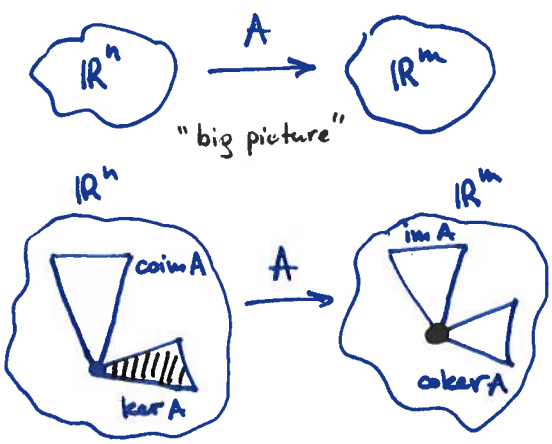
$x \in \ker(A) \iff Ax = 0 \iff (Ax)_i = 0, \ i = 1, \dots, m$

$(Ax)_i = \sum_{j=1}^n a_{ij} x_j = \underbrace{a^i}_{i^{th}\text{ row}} \cdot x$

Thus, $x \in \ker(A) \iff x \perp a^i$ for all $i = 1, \dots m$

But $\text{coim}(A) = \text{span}\{a^1 \dots a^m\}$, therefore $x \perp \text{coim}(A)$, and thus $x \in \ker(A) \iff x \in \text{coim}(A)^\perp$.

$\text{coker}(A) = \ker(A^T) = \text{coim}(A^T)^\perp = (\text{im}\, A)^\perp$ $\square$



maybe don't need this here

~~This result allows to derive an important characterization of compatible linear systems.~~

~~Q: When $Ax = b$ is compatible (i.e. has a solution)?~~

~~[We know the following criterion: $\text{rank}[A|b] = \text{rank}\, A$]~~

~~If and only if $b \in \text{im}(A) = \text{coker}(A)^\perp$ This alternative criterion is called Fredholm alternative~~

~~Fredholm Alternative: $Ax = b$ is compatible $\iff b \perp \text{coker}(A)$~~